



Calculating any term

Task A: Generating sequences using n^{th} terms

1) Use substitution to find the first five terms of these arithmetic sequences.

a) $5n + 3$

Term number (n)	1	2	3	4	5
Term	8	13	18	23	28

b) $0.05n + 0.3$

Term number (n)	1	2	3	4	5
Term	0.35	0.4	0.45	0.5	0.55

c) $3 - 0.7n$

Term number (n)	1	2	3	4	5
Term	2.3	1.6	0.9	0.2	-0.5

d) $-3 - 0.7n$

Term number (n)	1	2	3	4	5
Term	-3.7	-4.4	-5.1	-5.8	-6.5

2) Jun says:

The sequence $n + 3$ must go up in three's because it says, "+3"



Use substitution to find the first five terms and show Jun that he is mistaken.

Term number (n)	1	2	3	4	5
Term	4	5	6	7	8

$n = 1$

$n = 2$

$n = 3$

$n = 4$

$n = 5$

$(1) + 3 = 4$

$(2) + 3 = 5$

$(3) + 3 = 6$

$(4) + 3 = 7$

$(5) + 3 = 8$

The sequence begins at 4 and goes up 1 each time.



Task B: Using n^{th} terms to find any term

1) The first five terms of the arithmetic sequence $42 - 5n$ are 37, 32, 27, 22, and 17.

Jacob wants to find the 20th term.

37, 32, 27, 22, 17, 12, 7, 2, -2, -7, -12, -17, -22, -27, -32, -37, -42, -47, -52, -57. There it is. The 20th term is -57

His error was $2 - 5 = -3$

You might have said,
"Your method was inefficient.
If you know the n^{th} term use
substitution.

$$42 - 5(20) = 42 - 100 = -58$$



Spot Jacob's error and explain a better method to him.

2) Find the 20th, 100th and 1000th terms of the below sequences

a) $7n - 6$

$$n = 20$$

$$7(20) - 6 \\ = 140 - 6 = 134$$

$$n = 100$$

$$7(100) - 6 \\ = 700 - 6 = 694$$

$$n = 1000$$

$$7(1000) - 6 \\ = 7000 - 6 = 6994$$

b) $6 - 7n$

$$n = 20$$

$$6 - 7(20) \\ = 6 - 140 = -134$$

$$n = 100$$

$$6 - 7(100) \\ = 6 - 700 = -694$$

$$n = 1000$$

$$6 - 7(1000) \\ = 6 - 7000 = -6994$$

c) $0.06 - 0.7n$

$$n = 20$$

$$0.06 - 0.7(20) \\ = -13.94$$

$$n = 100$$

$$0.06 - 0.7(100) \\ = -69.94$$

$$n = 1000$$

$$0.06 - 0.7(1000) \\ = -699.94$$

d) $n^2 + 3n$

$$n = 20$$

$$(20)^2 + 3(20) \\ = 400 + 60 = 460$$

$$n = 100$$

$$(100)^2 + 3(100) \\ = 10\,000 + 300 = 10\,300$$

$$n = 1000$$

$$(1000)^2 + 3(1000) \\ = 1\,003\,000$$

e) $3n + 15 - 4n^2$

$$n = 20$$

$$3(20) + 15 - 4(20)^2 \\ = -1525$$

$$n = 100$$

$$3(100) + 15 - 4(100)^2 \\ = -39\,685$$

$$n = 1000$$

$$3(1000) + 15 - 4(1000)^2 \\ = -3\,996\,985$$



Task C: Comparing n^{th} terms of different sequences

Find the first five terms of each of these sequences and compare them:

- a) $3n$ 3, 6, 9, 12, 15, ...
- b) $3n + 2$ 5, 8, 11, 14, 17, ...
- c) $3n - 1$ 2, 5, 8, 11, 14, ...
- d) $3n + 0.7$ 3.7, 6.7, 9.7, 12.7, 15.7, ...
- e) $3n + \frac{1}{7}$ $3\frac{1}{7}, 6\frac{1}{7}, 9\frac{1}{7}, 12\frac{1}{7}, 15\frac{1}{7}, \dots$
- f) $-3n + 7$ 4, 1, -2, -5, -8, ...
- g) $2 - 3n$ -1, -4, -7, -10, -13 ...
- h) n^3 1, 8, 27, 64, 125, ...

Sequences a) to g) were all arithmetic. They all had a constant common difference.

a) to e) all had a common difference of +3. They were increasing sequences.

f) and g) had a common difference of -3, decreasing.

h) is not arithmetic.

